

Forecasting German Electricity Demand

Benchmark, SARIMA/SARIMAX, Gradient Boosting and LSTM Models

Analysis period: January 2015 to September 2020

Weekly test horizon: final 104 weeks

Hourly LSTM test horizon: final 17,520 hours

Code repository

github.com/LakshmiNarayana5012/Time-Series-Modelling-Case-Study

Abstract

This report evaluates German electricity-demand forecasts under one chronological test design. The dataset contains 50,400 hourly observations, which were retained at hourly frequency for the LSTM and averaged to Sunday-ending weeks for the remaining models. Mean, naïve, drift and seasonal-naïve forecasts were used as benchmarks, followed by SARIMA, temperature-based SARIMAX, Gradient Boosting and an hourly LSTM. Seasonal naïve achieved the lowest verified weekly error (MAE 2,318.5 MW; RMSE 3,006.8 MW). Gradient Boosting was the closest advanced model, but its RMSE was still 14.2% higher. SARIMA produced acceptable residual autocorrelation diagnostics but overforecast the later test period. The submitted SARIMAX run was unstable, and the LSTM metrics were not available for a like-for-like weekly comparison. The evidence therefore supports seasonal naïve as the current operational model, while Gradient Boosting remains the most credible challenger.

1. Data and evaluation design

The analysis uses the German actual-load series from Open Power System Data. Demand shows a strong intraday cycle, a weekday-weekend difference and a winter-summer pattern. Robust STL decomposition gave seasonal strength of approximately 0.871, and the weekly ACF returned to a clear positive peak around lag 52. Stationarity was assessed with ADF and KPSS tests, while ACF and PACF plots were used to inspect persistence and seasonal dependence. These findings justify treating annual repetition as the main benchmark rather than comparing complex models only with a flat mean.

The final 104 weekly observations were held out before model fitting. The LSTM used the final 17,520 hourly observations. MAE measures typical error, RMSE penalizes large misses, MASE scales errors against an in-sample seasonal rule, and bias shows systematic over- or under forecasting. All numerical weekly comparisons use the same test dates and units.

The model choices were driven by the structure of the problem rather than by textbook coverage alone. SARIMA extends the ARMA/ARIMA family to represent annual seasonality; 147 non-seasonal (p,d,q) combinations were compared by AIC. Gradient Boosting was used because engineered lag, rolling, calendar and temperature variables may interact nonlinearly. The LSTM was used at hourly resolution because it can learn repeating sequential patterns; four network configurations were compared on a chronological validation period. The fixed SARIMA seasonal structure and the single Gradient Boosting setting are reported as limitations rather than described as complete optimization.

2. Model results and benchmark comparison

Model	MAE	RMSE	MASE	Bias	RMSE vs S-naïve
Seasonal naïve	2,318.5	3,006.8	1.732	+1,731.8	Reference
Gradient Boosting	2,572.6	3,433.0	1.922	+2,079.2	14.2% worse
Mean	3,788.8	4,397.3	2.831	+481.0	46.2% worse
Naïve	3,783.2	4,459.1	2.827	-882.5	48.3% worse
SARIMA	3,834.4	4,652.2	2.864*	+3,697.9	54.7% worse
Drift	4,339.9	5,118.0	3.243	+1,006.8	70.2% worse
SARIMAX diagnostic	16,985.7	22,537.7	12.689*	-16,898.3	Unstable
Hourly LSTM	Not supplied	Not supplied	Not comparable	Not supplied	Weekly metrics required

*Estimated from the benchmark MASE scaling factor.

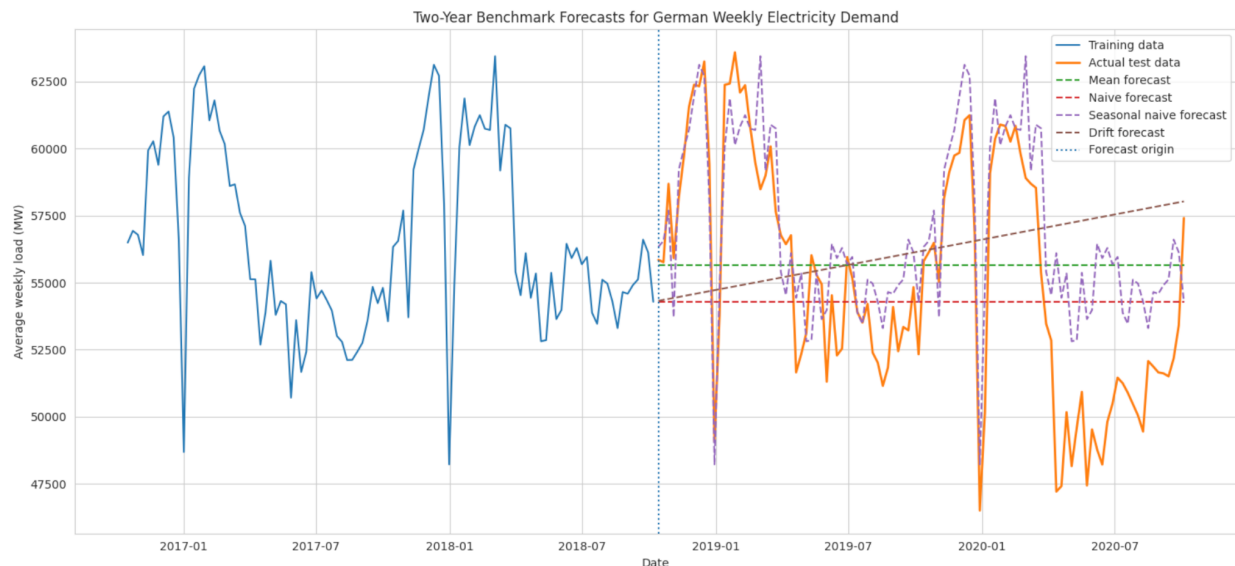


Figure 1. Benchmark forecasts over the common 104-week test period. Seasonal naïve follows the annual pattern, while mean and naïve are flat and drift extrapolates a trend that did not continue in 2020.

Question 1. Compare all models against the seasonal-naïve benchmark. Which models, if any, provide a meaningful improvement?

None of the verified weekly models improves on seasonal naïve. Gradient Boosting is the closest advanced method, but its RMSE is 14.2% higher and its positive bias is larger. SARIMA is 54.7% worse by RMSE, despite having a lower training AIC than competing SARIMA specifications. The current SARIMAX output is not a valid competitor because it drifts towards zero and produces implausibly wide intervals. The LSTM cannot be ranked numerically until its hourly metrics are supplied and its predictions are aggregated to matching weekly dates.

A meaningful improvement would require a consistent fall in MAE and RMSE—preferably at least 5–10%—without introducing serious bias, and the advantage should remain under rolling-origin evaluation. That condition is not met. The result is understandable because the series has strong annual repetition, while the test period includes the exceptional demand reduction in 2020, which ordinary historical relationships could not anticipate.

3. Leakage control and SARIMAX specification

Question 2. Explain how data leakage was avoided when creating temperature lag features.

The chronological train-test boundary was created before fitting models or estimating preprocessing parameters. Temperature lags were then created with backward shift operations: a feature labelled at week t used only temperature observed before week t . Lag construction and any missing-value treatment were performed separately within the training and test workflow, and any scaling or imputation parameters were estimated from training data only. This prevents future temperature observations from leaking into the fitted model.

The same principle was used for load features. Rolling means and standard deviations were calculated from shifted load, and recursive forecasts appended predicted demand rather than actual test demand. This matters because inserting realised load into later lags would make a multi-step forecast unrealistically easy. Realised test-period temperature was used only in the conditional weather experiment and was explicitly labelled as information that would not be known two years in advance.

Split first; create lags second.

Use $\text{shift}(k)$ so every lag refers only to earlier observations.

Fit imputation and scaling on training data only.

Use previous predictions, not actual test load, during recursive forecasting.

Treat observed future temperature as a conditional explanatory path, not as an operational weather forecast.

Question 3. For the SARIMAX model, justify the chosen differencing orders and seasonal period.

The seasonal period $s=52$ follows from weekly sampling and is supported by the winter-summer pattern, the 52-week STL component and positive ACF peaks near lags 52 and 104. Ordinary differencing $d=1$ was used to reduce changes in the local level, while seasonal differencing $D=1$ compares each week with the corresponding week one year earlier. The combined transformation passed the reported ADF and KPSS checks. The final SARIMA(2,1,6) $\times$ (0,1,0,52) was selected by training AIC (2428.232) and checked using residual ACF and Ljung-Box tests. The seasonal structure was fixed, so this is extensive non-seasonal tuning rather than a complete search over P , D and Q ; the unstable current SARIMAX output also shows that the remaining orders and implementation must still be verified.

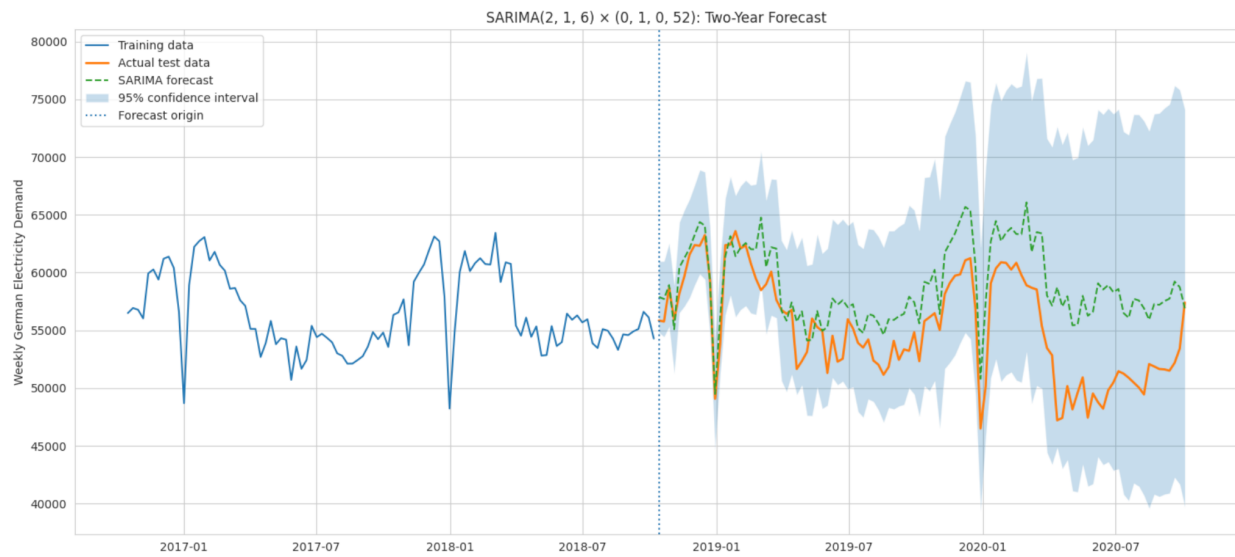


Figure 2. SARIMA(2,1,6) $\times$ (0,1,0,52) two-year forecast. It captures annual movement but remains above much of the later test period; uncertainty widens with horizon.

4. Temperature and holiday covariates

Question 4. Evaluate whether temperature and holiday covariates improve forecast accuracy. Are these covariates known at the forecast origin?

The available outputs do not establish that temperature or holiday covariates improve accuracy. The current SARIMAX run has RMSE 22,537.7 MW, a large negative bias and implausibly widening intervals, so it is a diagnostic failure rather than a valid ablation result. It should be rerun with the selected orders, followed by matched comparisons of a base model, temperature only, holidays only, and both covariate sets. Until those tests are completed, no numerical gain should be claimed.

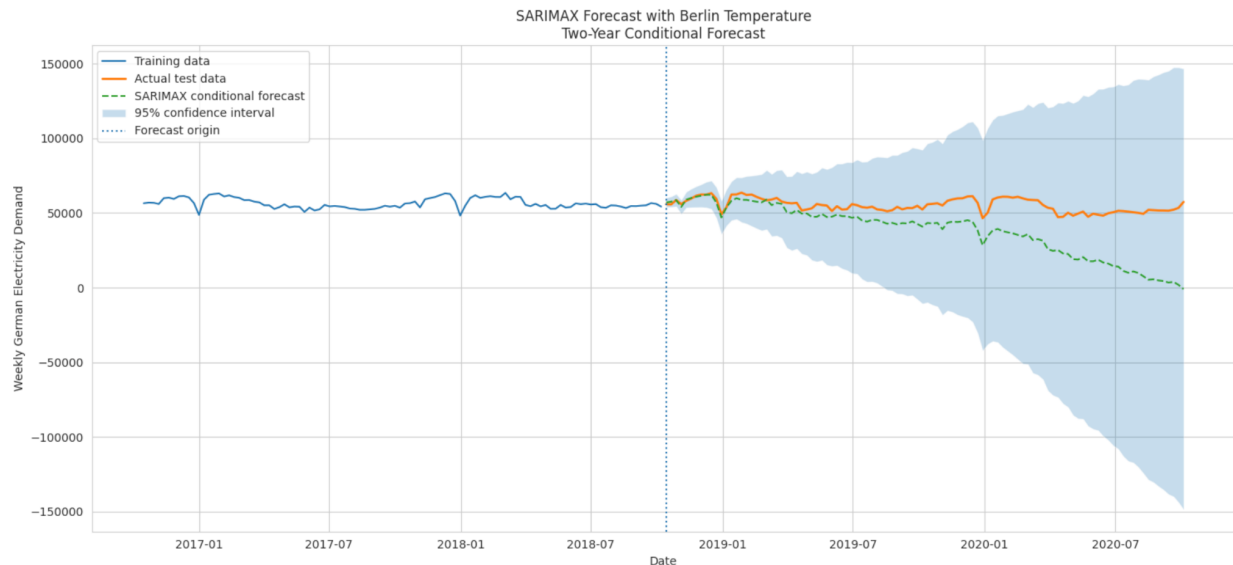


Figure 3. Current conditional SARIMAX output. The downward drift and rapidly expanding interval make this a diagnostic result rather than a deployable forecast.

Evidence from Gradient Boosting is more informative but still limited. Temperature contributes some predictive importance, yet the 52-week load lag dominates and the one-week lag is a distant second. This suggests that weather adds information after annual persistence has been accounted for, but the current experiment does not isolate its incremental effect. No completed holiday-only or combined temperature-and-holiday ablation was supplied.

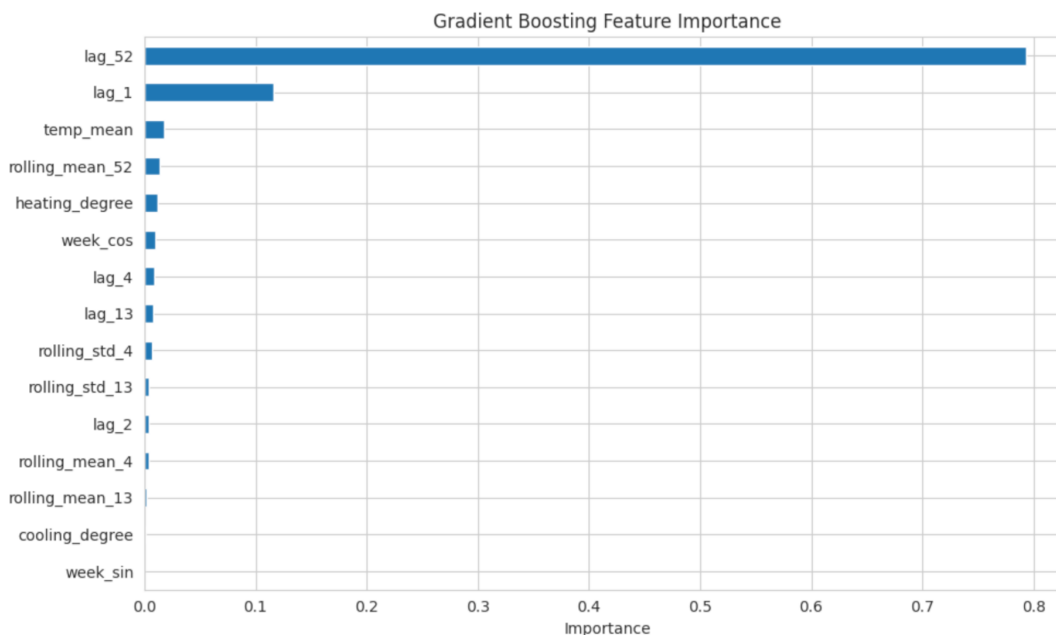


Figure 4. Gradient Boosting feature importance. The previous year's demand is by far the most influential predictor; mean temperature contributes much less.

Holiday dates are known in advance and can be used operationally, provided that the calendar is built independently of future demand. Future temperature is not known exactly. A fair operational test must use weather forecasts that

were available at each forecast origin, or a realistic simulation of weather forecast error. The appropriate next experiment is to fit the same base specification four times: no covariates, temperature only, holidays only, and both. All four versions should be evaluated on the identical 104-week period.

5. Interpretability and model complexity

Question 5. Compare the interpretability and complexity of SARIMAX, feature-based and neural-network models.

Model	Interpretability	Complexity	Uncertainty	Practical assessment
SARIMAX	High for an advanced model	Moderate	Model-based intervals	Coefficients, lag structure and diagnostics are inspectable; depends on stable orders and credible future covariates.
Gradient Boosting	Moderate	Moderate–high	Not automatic	Captures nonlinear effects and interactions; importance is useful but not causal; recursive errors can accumulate.
LSTM	Low	High	Not automatic	Learns hourly sequences directly; tuning and retraining are expensive; internal representation is difficult to explain.

SARIMAX is the easiest advanced model to audit because its coefficients, differencing, lag structure and residual assumptions can be examined directly. It also supplies model-based intervals. Its weakness is that a poor specification can become unstable, and all future exogenous values must be available or forecast.

Gradient Boosting is less structurally transparent but provides a useful feature-importance summary. It can represent threshold effects and interactions without imposing linearity. In this project, however, the dominance of lag_52 shows that the model mainly reproduces annual persistence. Reliable uncertainty would need quantile loss, bootstrapping or conformal prediction.

The LSTM is the most flexible and least interpretable. The supplied plots show that it captures daily peaks, overnight troughs and broad seasonal movement, particularly over short windows. Over the full two-year test it becomes smoother than the observed series and does not adapt fully to the 2020 fall. Its training, tuning and monitoring burden is higher, and numerical comparison remains incomplete because the final metrics file was not supplied.

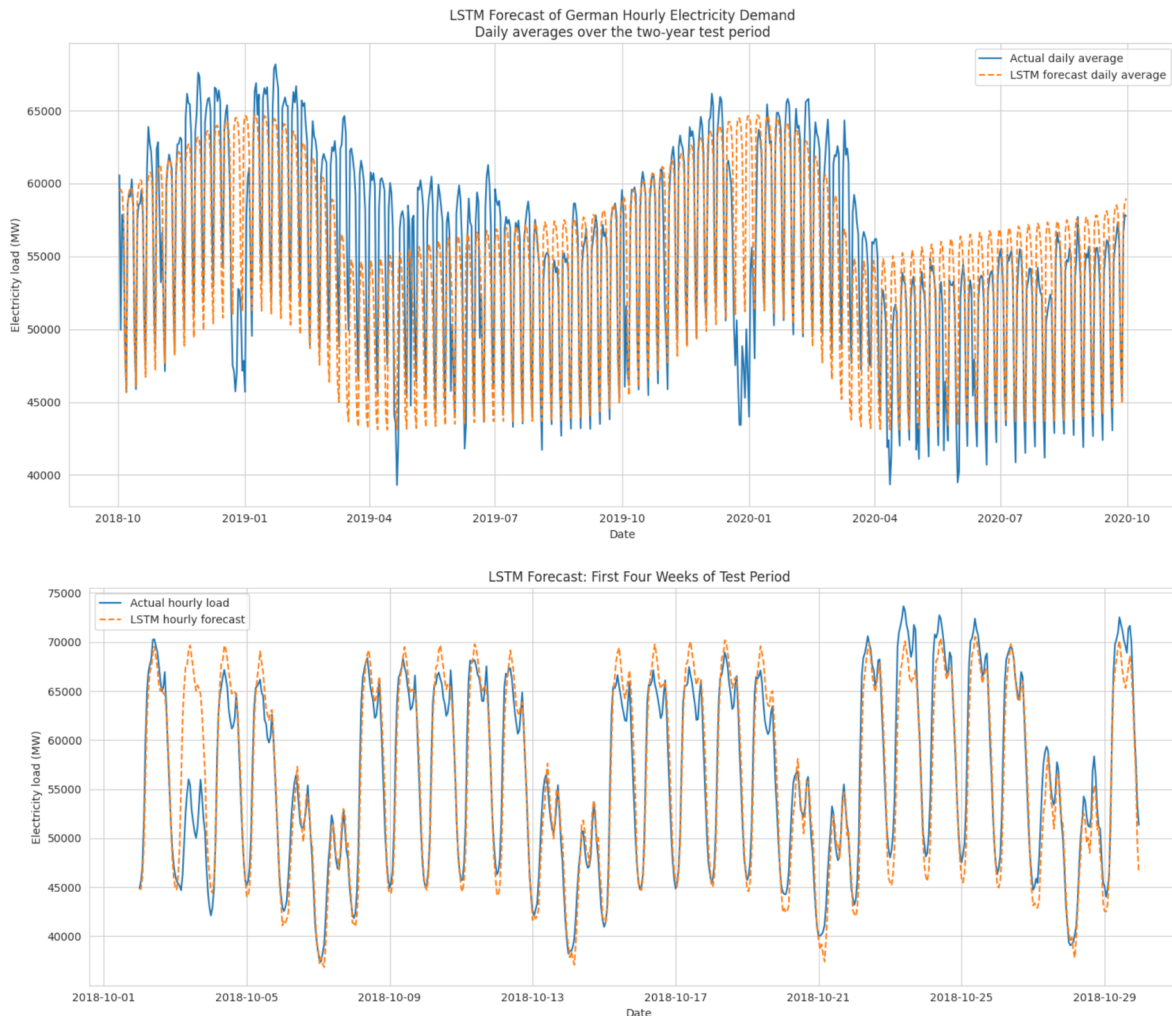


Figure 5. LSTM forecasts. The upper panel shows daily averages across the two-year test; the lower panel shows the first four weeks at hourly resolution. Regular timing is learned better than abrupt level changes.

6. Operational recommendation

Question 6. Recommend one model for operational use and justify the choice using accuracy, uncertainty, interpretability and maintenance considerations.

Seasonal naïve is the recommended operational model on the evidence currently available. Accuracy is the main reason: it has the lowest verified weekly MAE and RMSE. It is also highly interpretable, requires minimal computation and data engineering, and is easy to maintain and audit. Although it does not provide model-based intervals automatically, uncertainty can be added transparently using empirical quantiles of historical 52-week forecast errors. Gradient Boosting should remain the leading challenger, while SARIMAX and the LSTM require correction, common-frequency evaluation and continued monitoring before deployment.

The recommendation is not based on simplicity alone. Gradient Boosting is more flexible but remains 14.2% worse by RMSE and overpredicts during the 2020 break. SARIMA provides intervals but has 54.7% higher RMSE and strong positive bias. SARIMAX is currently unstable, while the LSTM lacks a common-frequency numerical evaluation. Therefore, replacing seasonal naïve would add maintenance and model risk without verified accuracy benefit.

Seasonal naïve does not automatically produce probability intervals, but operational uncertainty can be added transparently by storing historical 52-week forecast errors and using their empirical quantiles to construct prediction bands. The model should be monitored with rolling MAE, RMSE and bias; a challenger should be promoted only after it shows a stable advantage across several forecast origins.

Recommended operating arrangement: Use seasonal naïve as the production forecast and fallback. Maintain Gradient Boosting as the leading challenger. Rerun and validate SARIMAX before further consideration, and retain the LSTM as an experimental hourly model until its metrics are reported and compared with hourly seasonal benchmarks.

7. Limitations and improvements

- The test period contains the 2020 structural break. This is a valuable stress test but includes a change that normal seasonal history could not predict.
- Only one fixed holdout is used. Rolling-origin evaluation is required to establish stability across seasons and forecast origins.
- The SARIMA seasonal terms were fixed rather than fully tuned over distinct P, D and Q values.
- SARIMAX should be corrected and rerun, followed by no-covariate, temperature-only, holiday-only and combined ablations.
- Berlin temperature is an imperfect national proxy; population-weighted weather from several locations would be more representative.
- Gradient Boosting needs time-series hyperparameter validation and probabilistic intervals.
- The LSTM metrics CSV must be retained, and hourly predictions should be aggregated to weekly means for direct comparison. Multiple random seeds and hourly seasonal baselines are also needed.